

LAB SESSION 8

OPEN ENDED LAB

Objective

The objective of this lab session is to assess your understanding and application of Object-Oriented Programming (OOP) concepts, problem-solving skills, and coding techniques. You will be evaluated based on your ability to design and implement a system using OOP principles within the session.

Problem Statement:

Design a Library Management System using OOP.

The system should allow the addition of books and members, and should support issuing and returning books.

Concepts to Demonstrate:

- Encapsulation
- Inheritance
- Polymorphism
- Constructor usage
- Method Overriding
- Clean code and reusability

```

#include <iostream>
#include <vector>
#include <string>
using namespace std;

// ----- Book Class (Part of Library & Member) -----
class Book {
private:
    string title;
    string author;
    int bookID;

public:
    Book(string t, string a, int id) : title(t), author(a), bookID(id) {}

    void displayBook() const {
        cout << "Book ID: " << bookID << ", Title: " << title << ", Author: " << author << endl;
    }

    int getBookID() const {
        return bookID;
    }

    string getTitle() const {
        return title;
    }
};

// ----- Person Base Class -----
class Person {
protected:
    string name;
    int age;

public:
    Person(string n, int a) : name(n), age(a) {}

    virtual void displayInfo() const {
        cout << "Name: " << name << ", Age: " << age << endl;
    }

    virtual ~Person() {} // Virtual destructor for polymorphism
};

```

```

// ----- Member Class (Derived from Person) -----
class Member : public Person {
private:
    int memberID;
    vector<Book> borrowedBooks; // Composition: Member "has-a" list of Books

public:
    Member(string n, int a, int id) : Person(n, a), memberID(id) {}

    void borrowBook(const Book& book) {
        borrowedBooks.push_back(book);
        cout << name << " borrowed \"" << book.getTitle() << "\"" << endl;
    }

    void displayInfo() const override {
        Person::displayInfo();
        cout << "Member ID: " << memberID << endl;
        cout << "Borrowed Books:" << endl;
        for (const Book& book : borrowedBooks) {
            book.displayBook();
        }
    }
};

```

```

// ----- Library Class (Whole - has Books) -----
class Library {
private:
    vector<Book> books; // Composition: Library "has-a" list of Books
    vector<Member> members;

public:
    void addBook(const Book& book) {
        books.push_back(book);
    }

    void addMember(const Member& member) {
        members.push_back(member);
    }

    void displayBooks() const {
        cout << "\n--- Available Books in Library ---" << endl;
        for (const Book& book : books) {
            book.displayBook();
        }
    }

    void displayMembers() const {
        cout << "\n--- Library Members ---" << endl;
        for (const Member& member : members) {
            member.displayInfo();
            cout << endl;
        }
    }
};

// ----- Main Function -----
int main() {
    // Create Library
    Library lib;

    // Create Books
    Book book1("1984", "George Orwell", 101);
    Book book2("To Kill a Mockingbird", "Harper Lee", 102);
    Book book3("The Great Gatsby", "F. Scott Fitzgerald", 103);

    // Add Books to Library
    lib.addBook(book1);
    lib.addBook(book2);
    lib.addBook(book3);

    // Create Members
    Member m1("Ali", 22, 1);
    Member m2("Sara", 20, 2);

    // Members borrow books
    m1.borrowBook(book1);
    m2.borrowBook(book2);
    m2.borrowBook(book3);

    // Add Members to Library
    lib.addMember(m1);
    lib.addMember(m2);

    // Display Library Info
    lib.displayBooks();
    lib.displayMembers();

    return 0;
}

```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
```

Ali borrowed "1984"

Sara borrowed "To Kill a Mockingbird"

Sara borrowed "The Great Gatsby"

--- Available Books in Library ---

Book ID: 101, Title: 1984, Author: George Orwell

Book ID: 102, Title: To Kill a Mockingbird, Author: Harper Lee

Book ID: 103, Title: The Great Gatsby, Author: F. Scott Fitzgerald

--- Library Members ---

Name: Ali, Age: 22

Member ID: 1

Borrowed Books:

Book ID: 101, Title: 1984, Author: George Orwell

Name: Sara, Age: 20

Member ID: 2

Borrowed Books:

Book ID: 102, Title: To Kill a Mockingbird, Author: Harper Lee

Book ID: 103, Title: The Great Gatsby, Author: F. Scott Fitzgerald

PS C:\C++ PROGRAMS>